OLA Data Analyst Project

ChatGPT Prompt to Create Data

Please create a spreadsheet with 1 lac rows, for Bengaluru city. Give the following columns. The data will be for 1 month. use the following column -

1. Date
2. Time
3. Booking ID
4. Booking Status
5. Customer ID
6. Vehicle Type
   - Auto
   - Prime Plus
   - Prime Sedan
   - Mini
   - Bike
   - eBike
   - Prime SUV
7. Pickup Location (Create dummy location points Take any 50 areas from Bangalore)
8. Drop Location (Take from dummy pickup locations)
9. Avg VTAT (Time taken to arrive at the vehicle)
10. Avg CTAT (Time taken to arrive the Customer)
11. Cancelled Rides by Customer
12. Reason for cancelling by Customer

- Driver is not moving towards pickup location
- Driver asked to cancel
- AC is not working (Only for 4-wheelers)
- Change of plans
- Wrong Address

13. Cancelled Rides by Driver

- Personal & Car related issues
- Customer related issue
- The customer was coughing/sick
- More than permitted people in there

14. Incomplete Rides
15. Incomplete Rides Reason

- Customer Demand
- Vehicle Breakdown
- Other Issue

16. Booking Value
17. Ride Distance
18. Driver Ratings
19. Customer Rating

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Keep the overall booking status success for this data at 62%. If the booking status is successful, then only fare charge ratings, average VTAT, average CTAT, and other data will be there.
Make sure orders cancelled by customers should not be more than 7%
Make sure orders cancelled drivers should not be more than 18%

Also, increase the number of orders on weekends and match days. Keep match day by using the following dates.
keep incomplete rides less than 6%
Keep order value high on weekends

keep Booking ID with 10 digits starting with CNR and then
keep booking value under 500 value 70% keep orders
above 500 value 28% keep remaining orders above 1000

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SQL Questions:

1. Retrieve all successful bookings:
2. Find the average ride distance for each vehicle type:
3. Get the total number of cancelled rides by customers:
4. List the top 5 customers who booked the highest number of rides:
5. Get the number of rides cancelled by drivers due to personal and car-related issues:
6. Find the maximum and minimum driver ratings for Prime Sedan bookings:
7. Retrieve all rides where payment was made using UPI:
8. Find the average customer rating per vehicle type:
9. Calculate the total booking value of rides completed successfully:
10. List all incomplete rides along with the reason:

Power BI Questions:

1. Ride Volume Over Time
2. Booking Status Breakdown
3. Top 5 Vehicle Types by Ride Distance
4. Average Customer Ratings by Vehicle Type
5. cancelled Rides Reasons
6. Revenue by Payment Method
7. Top 5 Customers by Total Booking Value
8. Ride Distance Distribution Per Day
9. Driver Ratings Distribution
10. Customer vs. Driver Ratings

Data Columns

1. Date
2. Time
3. Booking_ID
4. Booking_Status
5. Customer_ID
6. Vehicle_Type
7. Pickup_Location
8. Drop_Location
9. V_TAT
10. C_TAT
11. cancelled_Rides_by_Customer
12. cancelled_Rides_by_Driver
13. Incomplete_Rides
14. Incomplete_Rides_Reason
15. Booking_Value
16. Payment_Method
17. Ride_Distance
18. Driver_Ratings
19. Customer_Rating

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SQL Answers:

1. **Retrieve all successful bookings:**
SELECT * FROM bookings WHERE Booking_Status = 'Success';

2. **Find the average ride distance for each vehicle type:**
SELECT Vehicle_Type, AVG(Ride_Distance) as avg_distance FROM bookings GROUP BY Vehicle_Type;

3. **Get the total number of cancelled rides by customers:**
SELECT COUNT(*) FROM bookings WHERE Booking_Status = 'cancelled by Customer';

4. **List the top 5 customers who booked the highest number of rides:**
SELECT Customer_ID, COUNT(Booking_ID) as total_rides FROM bookings GROUP BY Customer_ID ORDER BY total_rides DESC LIMIT 5;

5. **Get the number of rides cancelled by drivers due to personal and car-related issues:**
SELECT COUNT(*) FROM bookings WHERE cancelled_Rides_by_Driver = 'Personal & Car related issue';

6. **Find the maximum and minimum driver ratings for Prime Sedan bookings:** SELECT MAX(Driver_Ratings) as max_rating, MIN(Driver_Ratings) as min_rating FROM bookings WHERE Vehicle_Type = 'Prime Sedan';

7. **Retrieve all rides where payment was made using UPI:** SELECT * FROM bookings WHERE Payment_Method = 'UPI';

8. **Find the average customer rating per vehicle type:**
SELECT Vehicle_Type, AVG(Customer_Rating) as avg_customer_rating FROM bookings GROUP BY Vehicle_Type;

9. **Calculate the total booking value of rides completed successfully:**
SELECT SUM(Booking_Value) as total_successful_value FROM bookings WHERE Booking_Status = 'Success';

10. **List all incomplete rides along with the reason:**
SELECT Booking_ID, Incomplete_Rides_Reason FROM bookings WHERE Incomplete_Rides = 'Yes';

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Power BI Answers:

Segregation of the views:

1. **Overall**
- Ride Volume Over Time
- Booking Status Breakdown

2. **Vehicle Type**
- Top 5 Vehicle Types by Ride Distance

3. **Revenue**
- Revenue by Payment Method
- Top 5 Customers by Total Booking Value
- Ride Distance Distribution Per Day

4. **Cancellation**
- Cancelled Rides Reasons (Customer)
- cancelled Rides Reasons(Drivers)

5. **Ratings**
- Driver Ratings
- Customer Ratings

Answers:

1. **Ride Volume Over Time:** A time-series chart showing the number of rides per day/week.
2. **Booking Status Breakdown:** A pie or doughnut chart displaying the proportion of different booking statuses (success, cancelled by the customer, cancelled by the driver, etc.).
3. **Top 5 Vehicle Types by Ride Distance:** A bar chart ranking vehicle types based on the total distance covered.
4. **Average Customer Ratings by Vehicle Type:** A column chart showing the average customer ratings for different vehicle types.
5. **cancelled Rides Reasons:** A bar chart that highlights the common reasons for ride cancellations by customers and drivers.
6. **Revenue by Payment Method:** A stacked bar chart displaying total revenue based on payment methods (Cash, UPI, Credit Card, etc.).
7. **Top 5 Customers by Total Booking Value:** A leaderboard visual listing customers who have spent the most on bookings.
8. **Ride Distance Distribution Per Day:** A histogram or scatter plot showing the distribution of ride distances for different Dates.
9. **Driver Rating Distribution:** A box plot visualizing the spread of driver ratings for different vehicle types.

10. **Customer vs. Driver Ratings:** A scatter plot comparing customer and driver ratings for each completed ride, analyzing correlations.

SQL Questions & Answers

Create Database Ola;
Use Ola;

#1. Retrieve all successful bookings:
Create View Successful_Bookings As
SELECT * FROM bookings
WHERE Booking_Status = 'Success';

#2. Find the average ride distance for each vehicle type:
Create View ride_distance_for_each_vehicle As
SELECT Vehicle_Type, AVG(Ride_Distance) as
avg_distance FROM bookings GROUP BY
Vehicle_Type;

#3. Get the total number of cancelled rides by customers:
Create View cancelled_rides_by_customers As
SELECT COUNT(*) FROM bookings
WHERE Booking_Status = 'cancelled by Customer';

#4. List the top 5 customers who booked the highest number of rides:
Create View Top_5_Customers As
SELECT Customer_ID, COUNT(Booking_ID) as total_rides
FROM bookings
GROUP BY Customer_ID
ORDER BY total_rides DESC LIMIT 5;

#5. Get the number of rides cancelled by drivers due to personal and car-related issues:
Create View Rides_cancelled_by_Drivers_P_C_Issues As
SELECT COUNT(*) FROM bookings
WHERE cancelled_Rides_by_Driver = 'Personal & Car related issue';

#6. Find the maximum and minimum driver ratings for Prime Sedan bookings:

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```
Create View Max_Min_Driver_Rating As
SELECT MAX(Driver_Ratings) as max_rating,
MIN(Driver_Ratings) as min_rating
FROM bookings WHERE Vehicle_Type = 'Prime Sedan';
```

#7. Retrieve all rides where payment was made using UPI:

```
Create View UPI_Payment As
SELECT * FROM bookings
WHERE Payment_Method = 'UPI';
```

#8. Find the average customer rating per vehicle type:

```
Create View AVG_Cust_Rating As
SELECT Vehicle_Type, AVG(Customer_Rating) as avg_customer_rating
FROM bookings
GROUP BY Vehicle_Type;
```

#9. Calculate the total booking value of rides completed successfully:

```
Create View total_successful_ride_value As
SELECT SUM(Booking_Value) as total_successful_ride_value
FROM bookings
WHERE Booking_Status = 'Success';
```

#10. List all incomplete rides along with the reason:

```
Create View Incomplete_Rides_Reason As
SELECT Booking_ID, Incomplete_Rides_Reason
FROM bookings
WHERE Incomplete_Rides = 'Yes';
```

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Retrieve All Answers

#1. Retrieve all successful bookings:

Select * From Successful_Bookings;

Date	Time	Booking_ID	Booking_Status	Customer_ID	Vehicle_Type	Pickup_Location	Drop_Location	V_TAT	C_TAT
7/25/2024 22:20	22:20:00	CNR2940424040	Success	CID225428	Bike	Magadi Road	Varthur	203	30
7/30/2024 19:59	19:59:00	CNR2982357879	Success	CID270156	Prime SUV	Sahakar Nagar	Varthur	238	130
7/2/2024 9:02	9:02:00	CNR1797421769	Success	CID939555	Mini	Rajajinagar	Chamarajpet	252	80
7/13/2024 4:42	4:42:00	CNR8787177882	Success	CID802429	Mini	Kadugodi	Vijayanagar	231	90
7/23/2024 9:51	9:51:00	CNR3612067560	Success	CID476071	Bike	Tumkur Road	Whitefield	133	40
7/29/2024 23:33	23:33:00	CNR4787583516	Success	CID923404	Prime Plus	Hosur Road	Jayanagar	35	55
7/26/2024 4:03	4:03:00	CNR7943634301	Success	CID647026	Prime Plus	Kammanahalli	Rajajinagar	238	95
7/27/2024 13:18	13:18:00	CNR4524472111	Success	CID540929	Auto	Cox Town	Yelahanka	126	35
7/16/2024 9:54	9:54:00	CNR8181602032	Success	CID167642	Bike	Indiranagar	MG Road	70	95
7/2/2024 10:25	10:25:00	CNR8090918544	Success	CID640151	Bike	Magadi Road	HSR Layout	126	95
7/5/2024 23:42	23:42:00	CNR3196156650	Success	CID243275	Bike	Electronic City	Langford Town	140	40
7/9/2024 11:11	11:11:00	CNR9975925287	Success	CID162055	Prime SUV	Magadi Road	RT Nagar	42	30
7/12/2024 14:44	14:44:00	CNR1591113431	Success	CID902781	eBike	Koramangala	Sarjapur Road	245	70
7/11/2024 20:42	20:42:00	CNR3650331573	Success	CID217093	eBike	Basavanagudi	Hulimavu	84	25
7/8/2024 22:33	22:33:00	CNR6013805089	Success	CID817034	Prime Sedan	Padmanabhan...	Jayanagar	168	65

#2. Find the average ride distance for each vehicle type:

Select * from ride_distance_for_each_vehicle;

Vehicle_Type	avg(Ride_Distance)
Prime Sedan	15.6621
Bike	15.8755
Prime SUV	15.2950
eBike	15.7414
Mini	15.5033
Prime Plus	15.2206
Auto	6.2053

#3. Get the total number of cancelled rides by customers:

Select * from cancelled_rides_by_customers;

count(*)
4079

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#4. List the top 5 customers who booked the highest number of rides:

Select * from Top_5_Customers;

Customer_ID	total_rides
CID706246	3
CID688426	3
CID887797	3
CID219102	3
CID954071	3

#5. Get the number of rides cancelled by drivers due to personal and car-related issues:

Select * from Rides_cancelled_by_Drivers_P_C_Issues;

COUNT(*)
2500

#6. Find the maximum and minimum driver ratings for Prime Sedan bookings:

Select * from Max_Min_Driver_Rating;

MAX(Driver_Ratings)	MIN(Driver_Ratings)
5	3

#7. Retrieve all rides where payment was made using UPI:

Select * from UPI_Payment;

Date	Time	Booking_ID	Booking_Status	Customer_ID	Vehicle_Type	Pickup_Location	Drop_Location	V_TAT	C_TAT	Canceled_Rides_by_Cu
7/30/2024 19:59	19:59:00	CNR2982357879	Success	CID270156	Prime SUV	Sahakar Nagar	Varthur	238	130	NULL
7/13/2024 4:42	4:42:00	CNR8787177882	Success	CID802429	Mini	Kadugodi	Vijayanagar	231	90	NULL
7/27/2024 13:18	13:18:00	CNR4524472111	Success	CID540929	Auto	Cox Town	Yelahanka	126	35	NULL
7/16/2024 9:54	9:54:00	CNR8181602032	Success	CID167642	Bike	Indiranagar	MG Road	70	95	NULL
7/2/2024 10:25	10:25:00	CNR8090918544	Success	CID640151	Bike	Magadi Road	HSR Layout	126	95	NULL
7/9/2024 11:11	11:11:00	CNR9975925287	Success	CID162055	Prime SUV	Magadi Road	RT Nagar	42	30	NULL
7/19/2024 21:18	21:18:00	CNR4443921904	Success	CID654618	Mini	Tumkur Road	Koramangala	231	50	NULL
7/25/2024 3:44	3:44:00	CNR7194303296	Success	CID538245	Mini	Mysore Road	Hennur	175	50	NULL
7/15/2024 17:11	17:11:00	CNR6494005067	Success	CID805360	Auto	Yelahanka	Malleshwaram	84	60	NULL
7/14/2024 5:25	5:25:00	CNR7142279862	Success	CID378034	eBike	Yeshwanthpur	JP Nagar	210	45	NULL
7/3/2024 0:58	0:58:00	CNR5176704322	Success	CID296026	Prime Plus	KR Puram	Hennur	287	40	NULL
7/10/2024 21:56	21:56:00	CNR7547352327	Success	CID976231	Prime Plus	Hulimavu	Rajarajeshwa...	210	105	NULL
7/6/2024 15:02	15:02:00	CNR1568684278	Success	CID709612	Prime Plus	Bannerghatta ...	Majestic	42	90	NULL
7/17/2024 3:30	3:30:00	CNR1050003752	Success	CID993137	Bike	Chamarajpet	Shivajinagar	308	110	NULL
7/1/2024 2:45	2:45:00	CNR9758857830	Success	CID528642	Prime Plus	HSR Layout	Magadi Road	308	70	NULL

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#8. Find the average customer rating per vehicle type:
Select * from AVG_Cust_Rating;

Vehicle_Type	avg_customer_rating
Prime Sedan	3.9
Bike	3.9
Prime SUV	3.9
eBike	3.9
Mini	4
Prime Plus	3.9
Auto	4

#9. Calculate the total booking value of rides completed successfully:
Select * from total_successful_ride_value;

total_successful_ride_value
13762477

#10. List all incomplete rides along with the reason:
Select * from Incomplete_Rides_Reason;

Booking_ID	Incomplete_Rides_Reason
CNR5176704322	Customer Demand
CNR9312632867	Vehicle Breakdown
CNR7924302885	Customer Demand
CNR1640228587	Other Issue
CNR7623690602	Other Issue
CNR9590311980	Customer Demand
CNR5863244684	Customer Demand
CNR9526078867	Customer Demand
CNR7154043084	Customer Demand
CNR3193710797	Other Issue
CNR7073850950	Customer Demand
CNR9952584604	Customer Demand
CNR5433575259	Vehicle Breakdown
CNR3575066041	Vehicle Breakdown
CNR7537935962	Customer Demand
CNR2722435581	Vehicle Breakdown